

HENRY LEFEBVRE

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Buffalo, MN | US Citizen

EDUCATION

B.Sc., Honours Computer Science

Sep. 2022 - May 2026

McGill University | GPA: 3.85 / 4.0

Montreal, QC

Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Concurrency, Databases, Compilers, Software Design, Machine Learning, Computer Graphics

EXPERIENCE

Research Assistant - Physics-Based Graphics

Jan. 2025 - Sep. 2025

McGill University - Computer Graphics Lab

Montreal, QC

- Researched acceleration structures in **Python** for real-time collision detection on deformable 3D meshes
- Designed an algorithm for updating bounded deformation trees, improving bounding-volume tightness by up to 50% over baseline update methods for a ~10-15% simulation performance uplift
- Built an evaluation pipeline over **10,000 3D models** to measure bounding tightness and update cost

Software Developer - Autonomous Underwater Vehicle

Dec. 2023 - Jun. 2024

McGill University - Robotics

Montreal, QC

- Developed and tested **computer vision** systems for an autonomous robotic submarine
- Transformed live underwater camera feeds into maps of labeled objects in global coordinates

Software Developer - MAIS Kernel

Dec. 2023 - Jun. 2024

McGill AI Society

Montreal, QC

- Worked in a small team to build an AI document-analysis web app using **Django** in **Docker** for rental agreements

Data Science & Machine Learning Intern

Mar. 2023 - Sep. 2023

M2M Tech

Remote

- Analyzed financial datasets with **Python** and **pandas**, building interactive **Bokeh** visualizations of market trends

PROJECTS

rt - Ray Tracer

Jun. 2024 - Sep. 2024

- Built a parallel CPU **ray tracer** in **Rust** that renders scenes with millions of polygons, supporting .gltf and .obj mesh loading, high-resolution textures, and physically based materials
- Accelerated rendering with **BVH** traversal, quasi-Monte Carlo sampling, and a live progressive preview

ApartMatch - 1st Place Hack at MAIS Hacks 2023

Oct. 2023

- Won first place among 300+ hackers with a **machine learning**-powered apartment-hunting web app
- Ranked and filtered scraped listings with an AI image-captioning model to surface the best apartments

LangSonic - CNN Speech Classifier

Nov. 2023

- Designed and trained a **Convolutional Neural Network** in **TensorFlow** to classify spoken language from audio
- Built a **data pipeline** transforming 450,000 Mozilla Common Voice audio clips into normalized Mel spectrograms

SKILLS

Languages: Python, Rust, Java, C++, JavaScript, Kotlin, SQL, Bash

Frameworks/Libraries: NumPy, pandas, Django, Flask, FastAPI, TensorFlow

Tools: Git, Docker, AWS, Linux